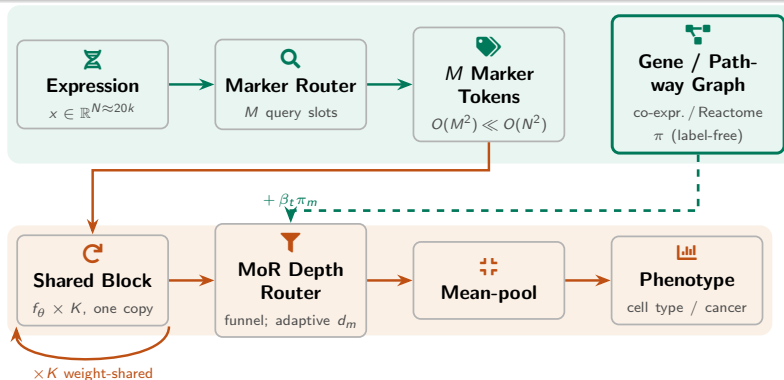


Problem & design: a biology-informed recursive transformer

Problem statement

Classify a cell/sample from $x \in \mathbb{R}^N$ ($N \approx 20k$ genes) *accurately & cheaply*. Two costs are our targets: **(T) token cost** — attention is $O(N^2)$ in genes; **(F) FLOP/depth cost** — a K -layer stack is $O(K)$ FLOPs and $O(K)$ params, spent uniformly on every token. **Goal: (T) $O(N^2) \rightarrow O(M^2)$, $M \ll N$; (F) $O(K) \rightarrow$ adaptive per-token depth at constant params** — guided by biology.



A marker selection: $N \rightarrow M$ tokens, $O(M^2)$ attention • **B** one shared block $\times K$ + **MoR** adaptive depth • **prior** π nudges **hubs** deeper (label-free)

Results: efficiency wins, accuracy is a wash (honest negatives)

Token reduction

macro-F1 vs # marker tokens M

	$M=16$	$M=64$	$M=128$	$M=256$
Tabula Muris	35.4	64.6	71.3	75.3
Prototype	62.5	90.2	94.8	97.2
Pancreas	43.9	53.0	58.4	57.7
Prostate	74.1	78.5	77.0	78.2

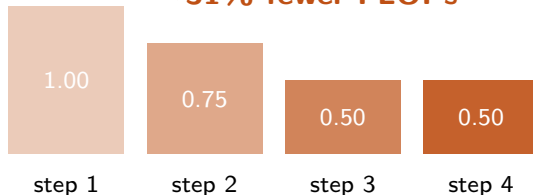
Vanilla vs Recursive vs MoR @ $K^*=4$

pancreas	Vanilla	Recursive	MoR
Params (unique)	299,136	74,784	75,168
FLOPs/sample	62.9 M	62.9 M	38.9 M
Compute saved	—	—	38%
macro-F1	54.1 ± 0.8	57.3 ± 5.8	54.2 ± 0.9
accuracy	92.4	91.8	91.1

Equal accuracy: MoR = $4 \times$ fewer params than Vanilla + 38% fewer FLOPs

Adaptive depth $\Rightarrow$ compute funnel

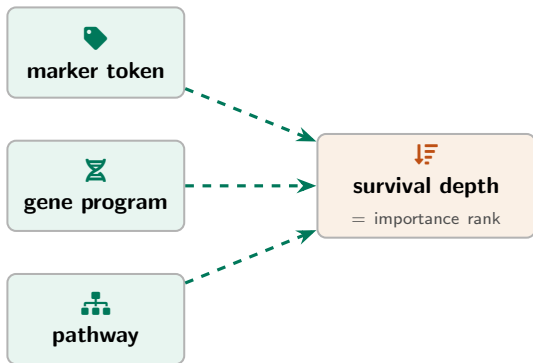
mean depth $2.75/4$
 $\sim 31\%$ fewer FLOPs



Verdicts (multi-seed, statistical)

- + Adaptive depth: $\sim 31\%$ compute saved at matched accuracy & params ($p < 0.001$).
- Biological prior vs random-graph: $\Delta F1 = +0.28$, $p = 0.871$, **0/10 significant**.
- Adaptive vs vanilla accuracy & calibration: **no consistent gain**.

Biologically meaningful token reduction — and where to push next



Each token $\approx$ a learned **signature / gene program** — a deployable **marker panel**, not a black box. A token's **recursion depth** is a compute-based **importance score**.

Negatives $\Rightarrow$ directions

- **Stronger injection:** prior is an additive, annealed-away bias $\rightarrow$ use **FiLM/gating** so centrality *modulates* routing per cell.
- **Better graph:** swap co-expression centrality for **directed GRNs / perturbation graphs** (causal, not correlational).
- **Right granularity:** route **pathway tokens** $\Rightarrow$ "depth per pathway" = compute a biological process needs.
- **Validate:** check survival-depth rankings against **known lineage / disease markers**.

Take-away: token reduction is the biologically meaningful lever that *works*; the bio prior is a promising direction, reported honestly.

Why a *learned* graph beats a fixed biological one

The router smooths expression along a gene graph A :

$$\tilde{x} = (1 - \lambda)x + \lambda Ax.$$

Fixed A (co-expression / Reactome):

- label-free $\Rightarrow \partial \mathcal{L} / \partial A = 0$ (edges never learn);
- annealed $\Rightarrow$ converged model $\equiv$ none;
- a degree-matched *shuffle* gives the same generic smoothing: $\mathbb{E}[\text{bio}] = \mathbb{E}[\text{random}]$ (FACT B).

Learned $A = \tilde{E}\tilde{E}^\top$ (rank r , embedding E):

$$\tilde{x} = (1 - \lambda)x + \lambda(x\tilde{E})\tilde{E}^\top, \quad \frac{\partial \mathcal{L}}{\partial E} \neq 0.$$

Edges are chosen to cut the loss. Low rank $\Rightarrow r$ latent **gene programs**; propagation denoises each gene toward its program's

One patient, two co-varying markers

Genes g_1, g_2 : weak, *noisy*, but co-vary with the label.

Learned graph learns $\tilde{E}_{g_1} \approx \tilde{E}_{g_2}$ (same program):

$$\tilde{x}_{g_1} \approx \frac{1}{2}(x_{g_1} + x_{g_2})$$

independent noise averages down ($\times \sqrt{2}$ **SNR**),
shared signal kept $\Rightarrow$ a **cleaner marker token**.

Random / fixed graph links g_1 to *unrelated* genes $\Rightarrow$
averages *in* noise $\Rightarrow$ over-smooths, no SNR gain.

On data (P-NET, 10 seeds): learned — none
= +1.9 F1 ($p=.043$); +5.7 on prostate. Beats
co-expression, Reactome *and* random — the first
mechanism that does.